

Basics of Communication Systems—Homework 1

Due date: 3/13

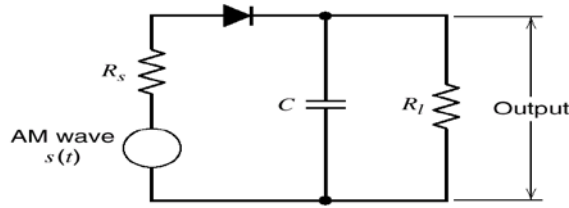
A. Double Sideband with Carrier

Let $m(t) = 0.1 + 3 \cos\left(2\pi f_m \frac{(t-1.5)}{25}\right) - 4 \sin\left(2\pi f_m \frac{(t-0.5)}{15}\right)$ be a continuous-time message signal, where $f_m = 100$. Modulate this signal by double-sideband amplitude modulation as

$$s(t) = A(1 + k_a \tilde{m}(t)) \cos(2\pi f_c t),$$

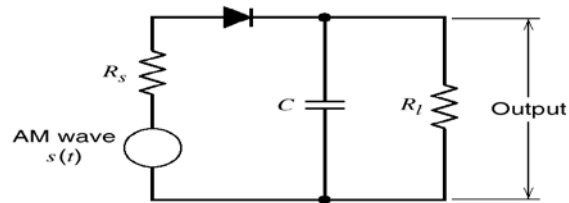
where A is the carrier amplitude, k_a is the modulation index, $\tilde{m}$ is the normalized signal such that $|\tilde{m}(t)| \leq 1$, and f_c is the carrier frequency.

1. Plot $m(t)$, the carrier, and $s(t)$ in the time domain.
2. The ideal envelop of this DBS-C signal is $e(t) = A(1 + k_a \tilde{m}(t))$. Obtain envelop $\hat{e}(t)$ via the following envelop detector circuit.



Plot and compare the detected envelop $\hat{e}(t)$ with the ideal $e(t)$.

3. Apply the following low-pass filter circuit to remove high-frequency components of $\hat{e}(t)$.



Plot and compare the output $\tilde{e}(t)$ of low-pass filter with $\hat{e}(t)$.

4. Recover the original message signal from the detected envelope using

$$\hat{m}(t) = \frac{1}{k_a} \left(\frac{\tilde{e}(t)}{A} - 1 \right).$$

Plot the original message and recovered signal on the same graph.

5. Discuss the effect of different values of k_a (e.g., 0.5, 1, 1.5) on distortion.

B. Double Sideband Suppressed Carrier

1. Plot $s_{SC}(t) = Am(t) \cos(2\pi f_c t)$ in the time domain.
2. Multiply $s_{SC}(t)$ by a local oscillator $\cos(2\pi f_c t + \theta)$, where θ is phase misalignment, and pass the result through the RC low-pass filter circuit. Plot the recovered message signal for $\theta = 0, \pi/6, \pi/3, \pi/2$ and explain how phase error affects the recovered signal amplitude.